

Template Week 3 – Hardware

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Assignment 3.1: Examine your phone

What processor is in your phone? Qualcomm SM7325 Snapdragon 778G 5G

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

How much RAM is in it? 8gb

How much storage does your phone have? 256 gb

What operating system is running on your phone? Android 14

Approximately how many applications do you have installed?

Which application do you use the most? YouTube

Can your phone be charged with what type of plug? Usb-c

Which I/O ports can you visually see on your phone? 3,5mm headphone jack

Assignment 3.2: Examine your laptop

What processor is in your laptop? Amd 5600H

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

How much RAM is in it? 16 gb ddr4

How much storage does your laptop have? 512

Which operating system is running on your laptop? Windows 10 home

Approximately how many applications do you have installed? 70

Which application do you use the most? Lebrewolf

Can your laptop be charged with what type of plug? A usb c

Which I/O ports can you visually see on your laptop? 2 usb-c 1 usb-a, 1 3,5 mm headphone jack.

Assignment 3.3: Power to the laptop

What is the input voltage? 240v

What is the output voltage? 20v

How many watts can your power adapter deliver? 90w

Is the input voltage AC or DC? ac

Is the output voltage AC or DC? dc

AC/DC what is that?

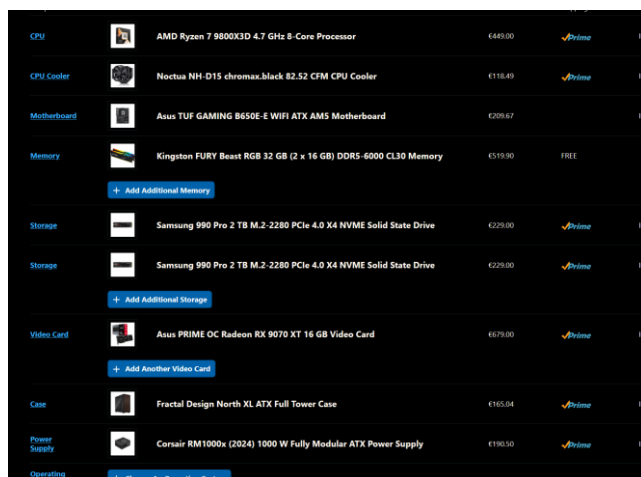
If you reverse the polarity of the output voltage, is that bad for your laptop? Yes that would fry it

You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that. Your device will take 15 watts due to built in safety mechanisms

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:

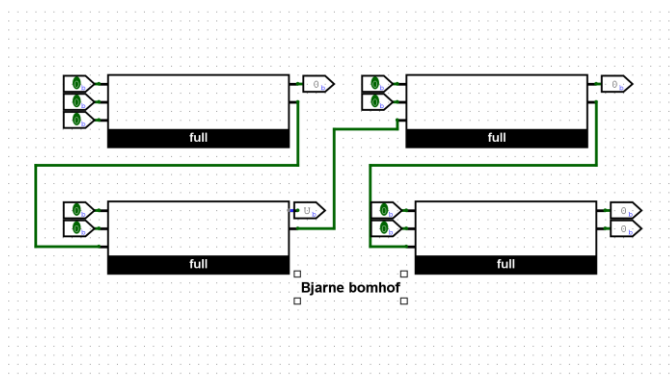
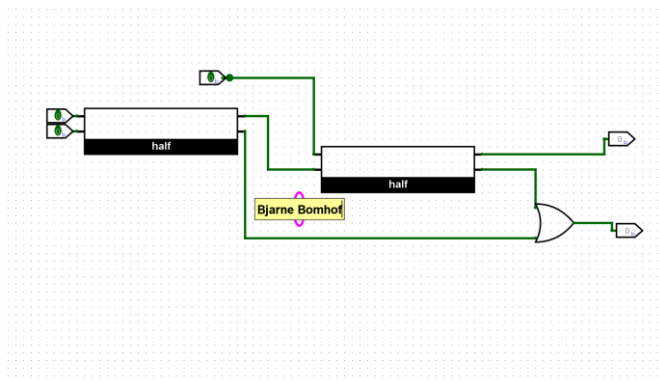
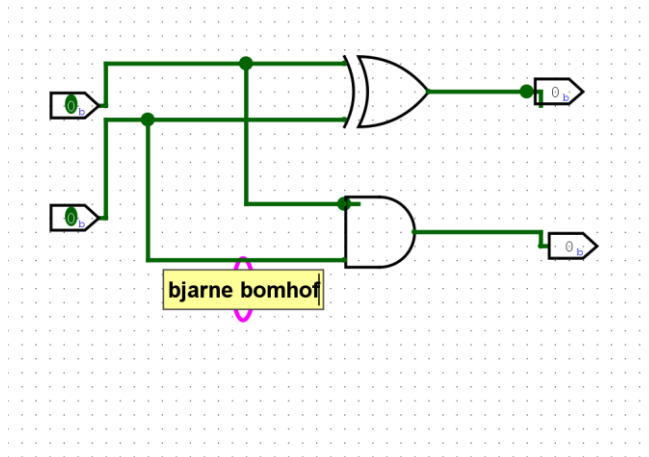
The highest spec computed with good Linux compatibility



Assignment 3.5: Adders

Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and also export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.



Ready? Save this file and export it as a pdf file with the name: [week3.pdf](#)